

EXPIREMENT-2

C Compilation Stages & Area of a Circle Program

- **Aim:** To write a C program to calculate the area of a circle and observe all four stages of compilation using the -save-temps GCC flag in Linux.
- **Tools Used:** Kali Linux Terminal, nano text editor, cc compiler.

C Source Code Used-

```
#include<stdio.h>    // Preprocessor directive to include builtin header file

#define PI 3.14      // Macro defining constant value for PI

int main(void)
{
    int radius;      // Variable to store radius
    float area;      // Variable to store calculated area

    printf("Enter the radius: "); // Display prompt
    scanf("%d", &radius);        // Read radius input from user

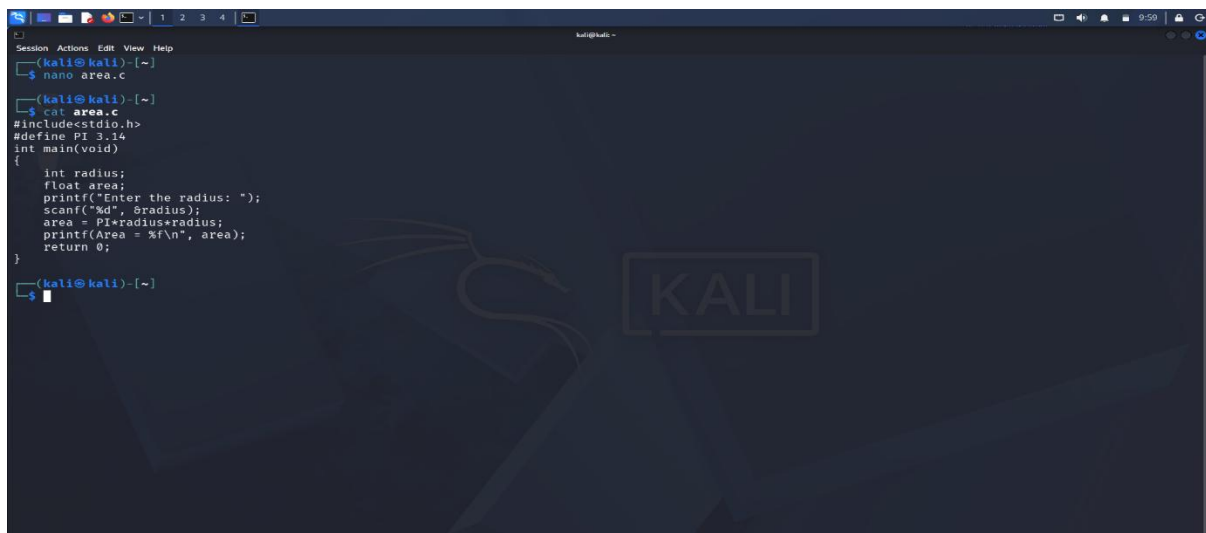
    area = PI * radius * radius; // Calculate area using formula  $\pi r^2$ 
    printf("Area = %f\n", area); // Output formatted result

    return 0;
}
```

Compilation Stage	Output File	Function / Description
1. Preprocessing	a-area.i	Expands #define macros (replaces PI with 3.14), removes comments, and includes header files (stdio.h).
2. Compilation	a-area.s	Translates preprocessed C code into architecture-specific Assembly instructions.
3. Assembly	a-area.o	Converts assembly language code into binary Machine Code (object file).
4. Linking	a.out	Combines object files with system C runtime libraries (for printf/scanf) to produce the final executable.

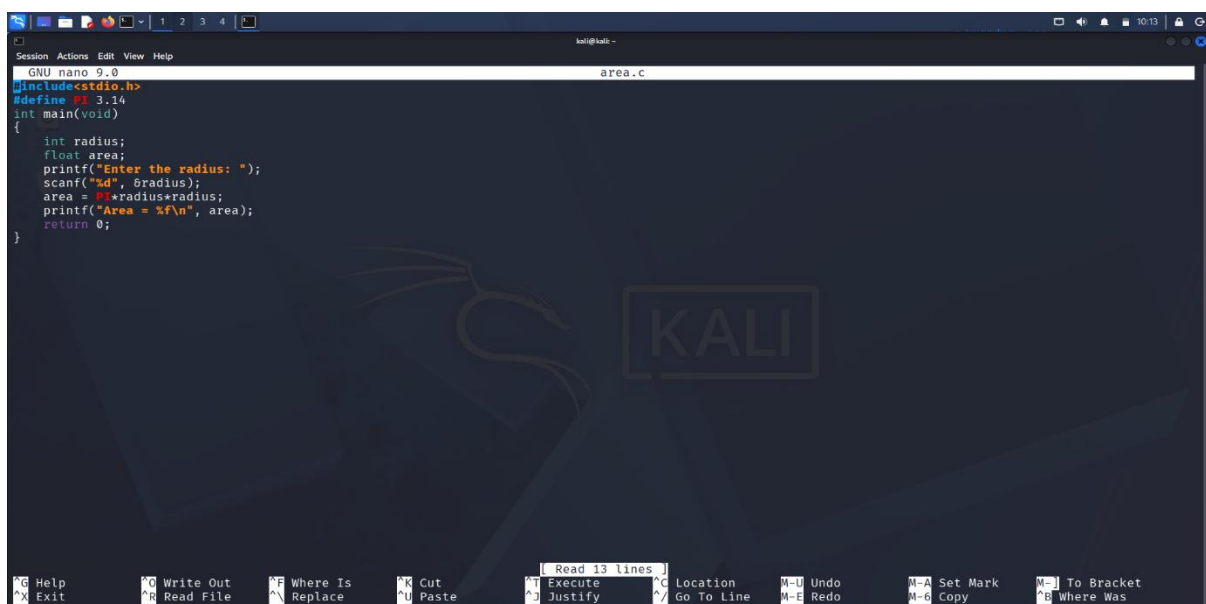
Command Sequence Used-

1. **Create/Open file:** nano area.c
2. **Compile with temporary files saved:** cc -save-temps area.c
3. **Verify generated files:** ls
4. **Execute program:** ./a.out



A terminal window on a Kali Linux system. The user has created a file named area.c using the nano text editor. The code inside area.c is as follows:

```
(kali@kali)-[~]  
$ nano area.c  
  
(kali@kali)-[~]  
$ cat area.c  
#include<stdio.h>  
#define PI 3.14  
int main(void)  
{  
    int radius;  
    float area;  
    printf("Enter the radius: ");  
    scanf("%d", &radius);  
    area = PI*radius*radius;  
    printf("Area = %f\n", area);  
    return 0;  
}
```



A terminal window showing the contents of the file area.c. The code is the same as in the previous screenshot. The terminal window title is "GNU nano 9.0 area.c".

```
GNU nano 9.0 area.c  
#include<stdio.h>  
#define PI 3.14  
int main(void)  
{  
    int radius;  
    float area;  
    printf("Enter the radius: ");  
    scanf("%d", &radius);  
    area = PI*radius*radius;  
    printf("Area = %f\n", area);  
    return 0;  
}
```

```
(kali@kali)-[~]
$ nano area.c

(kali@kali)-[~]
$ cc -save-temps area.c

(kali@kali)-[~]
$ ls
a-area.i  a-area.s  area.c  Desktop  Downloads  mkdir  Pictures  Public  Templates
a-area.o  a.out     demo    Documents help       Music    Projects sample.c Videos

(kali@kali)-[~]
$ ./a.out
Enter the radius: 5
Area = 78.500000

(kali@kali)-[~]
$
```

1. **Compilation success:** Command `cc -save-temps area.c` executed without error.
2. **Generated files:** `ls` output showing `a-area.i`, `a-area.s`, `a-area.o`, and `a.out`.
3. **Execution output:**

Enter the radius: 5

Area = 78.500000

Conclusion-

"The C program to calculate the area of a circle was written, compiled, and executed successfully. All four intermediate stages of compilation (Preprocessing, Compilation, Assembly, and Linking) were verified by inspecting the generated -save-temps temporary files."